EV Market Performance and Strategic Dashboard Insights

Data Visualization, Business Question Mapping, and Multi-Persona Decision Support System

Date	28 October 2026	**Lead Analyst**	Naveen Patidar
Project Title	EV Market Performance & Analytics	**Technical Tool**	Tableau Desktop / Business Analytics
Evaluation Track	Capstone Milestone 6.1 Report	**Maximum Marks**	5 Marks

1. Industry Context & Executive Summary

The electric vehicle (EV) industry has emerged as one of the fastest-growing sectors in the global automotive landscape, propelled by rigorous carbon neutrality commitments, government incentive frameworks, and breakthroughs in battery chemistry. Contemporary vehicle lineups span hatchbacks, sedans, full-size SUVs, electric buses, haulage trucks, and micro-mobility platforms. To navigate intensifying competitive dynamics, automotive stakeholders require interactive intelligence syntheses that connect vehicle sales, charging efficiency, battery lifecycles, and environmental dividends.

2. Stakeholder User Personas & Strategic Requirements

Emily Johnson

EV Market Strategy Director

Focus: Evaluating category sales contributions, driving range dispersion, and portfolio shifts.

Decision Need: Interactive intelligence to identify high-growth market segments and target future model introductions across key regions.

David Chen

Automotive BI Manager

Focus: Monitoring manufacturer market share, regional production hubs, and brand satisfaction indices.

Decision Need: Benchmarking original equipment manufacturer (OEM) capabilities to provide strategic guidance on market entry and positioning.

Dr. Sarah Williams

Sustainable Mobility Researcher

Focus: Chemistry comparison (LFP, NMC, NCA, Solid-State), degradation curves, and net CO2 offset.

Decision Need: Empirical modeling linking technological innovations with lifecycle durability and environmental policy efficacy.

3. Visualization Engineering & Dashboard Architecture

Visualization development translates high-dimensional automotive telemetry and market data into cohesive graphical representations. By establishing systematic relationships between visual channels—such as spatial coordinates, color palettes, and faceted views—the designed Tableau system addresses core operational questions while allowing seamless drill-down from macro-level regional summaries to micro-level battery degradation metrics.

4. Master Dashboard Architecture & Tableau Implementation

The master dashboard `Dashboard 1` integrates eight analytical worksheets into an interactive executive decision cockpit. Built by Naveen Patidar, this single-pane interface correlates high-level annual sales volume across vehicle classes with granular engineering parameters including battery degradation, cycle life, and geographic production concentration.



Figure 1.0: Naveen Patidar's Multi-Faceted Tableau Dashboard 1 showcasing 8 coordinated visual worksheets.

5. Architectural Layout Breakdown

Top Band: Volume & Ratings

Worksheets: Annual Sales by Vehicle Category, Annual Sales by OEM, and Customer Rating vs. Driving Range scatter layout.

Mid Band: Market Geography

Worksheets: S2_Regional_Market_Share (radial pie), S2_Battery_by_Manufacturer (area chart), and S2_Manufacturing_Map (geographic plot).

Lower Band: Technical Depth

Worksheets: S3_Battery_Tech_Range (clustered bar), S3_Battery_Lifecycle (degradation plot), and S3_CO2_Impact (environmental bar chart).

Key Architectural Accomplishment: Real-time cross-filtering enables Emily Johnson, David Chen, and Dr. Sarah Williams to filter by drivetrain, battery chemistry, and manufacturer directly from the Tableau workbook sheets without resetting analysis parameters.

6. Core Business Questions & Visual Analytics (Part I: Questions 1 to 4)

Question 1: Which vehicle categories and countries lead global sales and manufacturing output?

Target Worksheet: Top-Left Horizontal Bar Chart & S2_Manufacturing_Map | Persona: Emily Johnson & David Chen

Visual Analysis: Sedans and Hatchbacks dominate total global delivery volume, each crossing 75M cumulative units, followed closely by Micro City Cars (~58M) and Luxury Sedans (~59M). The spatial manufacturing map identifies primary assembly cluster hubs in China, the United States, Japan, South Korea, and Germany, demonstrating that Asian and North American clusters retain production leadership.

Question 2: How do major manufacturers perform across sales volume and customer satisfaction?

Target Worksheet: Annual Sales Units by Manufacturer & S1_Rating_vs_Range | Persona: David Chen

Visual Analysis: Evaluating OEM performance across AmpereMobility, AxleGroup, ChargeEV, CircuitDynamics, DriveAuto, TorqueWorks, and ZapGroup reveals consistent unit outputs clustered between 2.5M and 4.0M units annually per enterprise. Cross-referencing against customer feedback shows positive rating clustering between 3.5 and 4.8 stars, highlighting that scale does not compromise user sentiment.

Question 3: What is the driving range capability across different vehicle categories?

Target Worksheet: S3_Battery_Tech_Range & Driving Range Distribution | Persona: Emily Johnson

Visual Analysis: Driving range exhibits sharp stratification: Luxury Sedans and Sports Cars achieve top performance medians between 480 km and 540 km. Mainstream Sedans and Crossovers form a strong middle tier between 360 km and 440 km, whereas Micro City Cars and Two-Wheeler Scooters occupy the lower bound (180 km to 240 km), reflecting battery pack size constraints.

Question 4: How does 3-year retained economic value compare across vehicle categories?

Target Worksheet: Retained Value 3yr Valuation Analysis | Persona: David Chen

Visual Analysis: Commercial and heavy-duty electric vehicles retain significant absolute residual capital: Electric Buses lead with an average retained valuation of $14,512,007, followed by Electric Trucks ($8.8M) and Sports Cars ($8.6M). Mass-market passenger vehicles (Sedans, Hatchbacks, Micro Cars) depreciate at standard automotive trajectories, sustaining residual values between $1.6M and $2.8M per fleet block.

7. Core Business Questions & Visual Analytics (Part II: Questions 5 to 8)

Question 5: How have annual sales trends evolved across model years by category?

Target Worksheet: Sales Trend Over Time (Multi-Line Chart) | Persona: Emily Johnson

Visual Analysis: Longitudinal tracking from model year 2017 to 2027 shows strong volume stability across Two-Wheeler Scooters and Sedans (tracking near 20M to 25M annual segment runs). While high-volume categories maintain flat-to-modest growth, specialized platforms like Electric Trucks and Commercial Heavy Haulers remain below 5M units, indicating significant unpenetrated growth runway.

Question 6: Which battery chemistries lead market adoption, and what are their cycle lifespans?

Target Worksheet: S3_Battery_Tech_Range & S3_Battery_Lifecycle | Persona: Dr. Sarah Williams

Visual Analysis: NMC and LFP represent the highest annual manufacturing volume due to optimized energy densities and proven cost structures. However, lifecycle plots demonstrate that Solid-State battery cells achieve superior longevity (exceeding 3,000 cycles with degradation under 5%), while standard LFP and NCA chemistries stabilize between 1,600 and 2,500 cycles with 10% to 15% degradation.

Question 7: What is the operational trade-off between charging speed and energy efficiency?

Target Worksheet: Charging vs. Efficiency (DC kW vs. Wh/km Scatter) | Persona: Dr. Sarah Williams

Visual Analysis: The scatter distribution highlights two distinct operational paradigms: Passenger sedans, hatchbacks, and micro vehicles cluster tightly in the high-efficiency zone (120 to 250 Wh/km) with moderate DC fast charging (50 to 150 kW). Conversely, electric heavy commercial haulers and transit buses cluster in the upper right quadrant (>1,000 Wh/km and 250 to 350+ kW charging), demanding mega-watt charging infrastructure.

Question 8: How much net CO2 is saved annually across global market regions and drivetrains?

Target Worksheet: S3_CO2_Impact & Regional Environmental Impact Map | Persona: Dr. Sarah Williams

Visual Analysis: Pickup Trucks, Electric Trucks, and Electric Buses contribute the highest cumulative carbon savings per deployed unit—saving between 40,000 and 50,000 metric tons of CO2 per year. Regionally, Asia-Pacific, Europe, and North America represent the highest gross emissions avoidance due to high renewable grid integration and high electric bus transit fleet turnover.

Strategic Conclusion & Evaluation Summary: Through this Tableau implementation, Naveen Patidar has established a unified analytical framework that directly satisfies the strategic oversight needs of Emily Johnson, the competitive intelligence requirements of David Chen, and the technological sustainability research goals of Dr. Sarah Williams.